

ClubTime! Relational Schema

User(userId [PK], firstName, lastName, DoB, joinDate, passwordHash, bannedByAdminID [FK -> Admin])

UserEmail(userId [PK, FK -> User], email [PK])

UserPhone(userId [PK, FK -> User], phone [PK])

Admin(adminId [PK], firstName, lastName, email)

Clubs(clubId [PK], clubName, clubStatus, clubFocus, lastActivityDate)

Club_Manager(userId [PK, FK -> User], startDate)

SocialMediaManager(userId [PK, FK -> Club_Manager])

EventCoordinator(userId [PK, FK -> Club_Manager])

ChatModerator(userId [PK, FK -> Club_Manager])

ClubFormationRequest(requestId [PK], userId [FK -> User], reviewedByAdminId [FK -> Admin], proposedClubName, club_Intention, proposedClubActivities, status, requestDate)

ClubManagement(clubId [PK, FK -> Clubs], userId [PK, FK -> Club_Manager])

Membership(userId [PK, FK -> User], clubId [PK, FK -> Clubs], dateJoined)

AdminManagesClub(adminId [PK, FK -> Admin], clubId [PK, FK -> Clubs])

BroadcastAuthors(broadcastId [PK, FK -> Broadcast], userId [PK, FK -> Club_Manager])

Broadcast(broadcastId [PK], broadcastTimestamp, clubId [FK -> Clubs])

EventUpdate(broadcastId [PK, FK -> Broadcast], eventDate, eventLocation, eventDescription)

Poll(broadcastId [PK, FK -> Broadcast], pollQuestion, pollCloseDate)

Announcement(broadcastId [PK, FK -> Broadcast], announcementText)

Notification(notificationId [PK], content, isRead, sentTimestamp, broadcastId [FK -> Broadcast], userId [FK -> User])

PostThread(postId [PK], content, createdDate, authorUserId [FK -> User])

Interaction(interactionId [PK], interactionType, content, timestamp, initiatorUserId [FK -> User], targetPostId [FK -> PostThread], recipientUserId [FK -> User])

Documentation:

Our relational schema is **BCNF compliant**. BCNF Normalization was achieved by design.

- Multivalued attributes from an entity are split into individual tables → avoid redundancy and is compliant with 1NF
- ISA subclasses utilize parent entity's PK → no redundancy across children classes.
- Junction tables were created to represent M:N (many-to-many) relationships; these tables possess composite PKs (*since PK is the combination of both entities' FKs, and all other attributes depend on this combination, not on either FK alone*)
- Every non-key attribute is dependent on a whole key; there are no non-key attributes determining any other attribute.
- There are no partial or transitive dependencies present across the relational schema.